

Computing intracoronary blood flow rate under incomplete boundary conditions: Combing coronary anatomy and fractional flow reserve

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ABSTRACT

Background: Accurate measurement of intracoronary blood flow rate is of great significance for the diagnosis of ischemic heart disease (IHD). Computational fluid dynamic (CFD) method, combining coronary angiography images and fractional flow reserve (FFR), provides a new way to calculate the mean flow rate. However, due to the incomplete boundary conditions obtained by FFR, side branches were ignored which was likely to have a significant impact on the accuracy. In this paper, a novel CFD based method for calculating the mean intracoronary flow rate under incomplete pressure boundary conditions was proposed, in order to improve the accuracy by including the side branches.

Methods: A pressure-flow curve based flow resistance model was employed to model resistance of the epicardial arteries. A series of steady flow simulations were performed to extract the parameters of the flow resistance model, which implicitly specified constraints for splitting flow between branches and thus enabled the mean intracoronary blood flow rate to be calculated in two or more branches under incomplete pressure boundary conditions. Simulation experiments were designed to validate the proposed method in both idealized and reconstructed 3D models of coronary branches, and the impact of the assumed coefficient of the Murray's Law for splitting flow between branches was also investigated.

Results: The mean percentage error of the proposed method was $+2.05\% \pm 0.04\%$ for idealized models and $+2.24\% \pm 0.01\%$ for reconstructed models, and it was much lower than that of the method ignoring side branches ($+38.48\% \pm 10.45\%$ for idealized models and $+30.54\% \pm 6.12\%$ for reconstructed models). When the assumed coefficient of the Murray's Law was inconsistent with the real blood flow condition, the percentage errors still maintained less than about 3.00%.

Conclusions: The proposed method provided an easy and accurate way to measure the mean intracoronary flow rate and would facilitate the accurate diagnosis of IHD.

Abbreviations

IHD	ischemic heart disease
CFD	Computational fluid dynamic
FFR	fractional flow reserve
iFR	instantaneous wave-free ratio

QFR	quantitative flow ratio
TIMI	Thrombolysis in Myocardial Infarction
APV	average peak velocity

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1. Introduction

Ischemic heart disease (IHD) is the single largest cause of death globally [1]. The underlying pathophysiology of IHD is complex involving various coronary compartments (epicardial vessels and/or microcirculation) with different mechanisms (anatomical atherosclerotic and/or functional alterations) [2]. Accurate measurement of intracoronary blood flow rate is of great significance for the diagnosis of IHD: firstly, it improves the identification of the hemodynamic severity of epicardial coronary stenoses and optimize decision making in interventional cardiology; secondly, it provides a way to assess the coronary microvascular disease which cannot be achieved by pressure-derived physiological indices such as fractional flow reserve (FFR) or instantaneous wave-free ratio (iFR) [3–6]; thirdly, it would facilitate the study of hemodynamic mechanisms of atherosclerosis by providing realistic patient-specific boundary conditions [7–9].

However, measurement of intracoronary flow is quite challenging. TIMI (Thrombolysis in Myocardial Infarction) frame count is a simple, inexpensive and widely used method to assess the coronary blood flow [10]. It was defined as the number of cine frames required for contrast to reach a standardized distal coronary landmark during coronary angiography. However, TIMI frame count was a semi-quantitative method, and the rate of filling of the contrast agent was not equivalent to the rate of blood flow, especially during the pulsating condition. Intracoronary wires equipped with Doppler sensor allowed measuring the velocities of red cells and the average peak velocity (APV) was quantitated [11]. The APV, as a surrogate of flow rate, was related to the flow profiles, which would be greatly affected by the insertion of Doppler sensor. The sensor angulation relative to the flow vector also had great impact on the measured velocity [12]. Moreover, it was challenging to maintain the Doppler ultrasound beam in a stable position within the narrow flow field as the heart beats [12]. Thermodilution technique provided another way to estimate the mean flow rate via the saline bolus transiting time [13]. However, it was affected by the quality of the injections and coronary flow rate (especially at high flow rates) [12]. Moreover, head-to-head comparison demonstrated that only moderate correlation ($\rho = -0.36$, $p = 0.002$ at baseline and $\rho = -0.41$, $p = 0.0003$ at hyperemia) was observed between simultaneously measurements of the transit mean time using thermodilution and the average peak velocity using Doppler sensor [14]. Recently, continuous thermodilution method was used to measure the absolute flow rate in coronary arteries [15]. By continuous infusing of saline with a known rate and temperature, flow rate could be measured from the mixing temperature of the blood [16]. Proper mixing was the determinant for this technique and to guarantee adequate mixing, the infusion catheter was required to be placed in section of artery with no side-branches within 3 cm distal to the infusion site, limiting the utilization of this method [16,17].

Recently, image-based computational fluid dynamic (CFD) method was introduced to assess coronary physiology. Since blood flow was governed by the mass conservation and Navier-Stokes equations, the pressure or flow rate can be derived from each other with appropriate CFD model (3D model and proper boundary conditions). The most successful clinical application of these CFD based methods was Quantitative flow ratio (QFR). With the reconstructed 3D model of the epicardial coronary artery from coronary angiography images, the pressure distribution could be calculated by using advanced CFD models [18,19]. However, intracoronary blood flow could not be obtained by QFR due to that only anatomy information was used. In order to assess intracoronary blood flow, Morris et al. proposed a novel CFD based method which combined anatomy and functional information [20]. Morris et al. firstly reconstructed 3D model of the epicardial coronary artery from multi-view coronary angiography images; then, with the measured proximal and distal mean pressure during FFR measurement, the corresponding mean flow rate was calculated by using CFD method [20]. *In vitro* experiment demonstrated that the CFD based approach was more accurate and reproducible than the Doppler sensor [20]. Since

measurement of intracoronary pressure is simple, accurate and broadly applied in catheterization laboratory, this new approach holds great potential for clinical utilization. However, routine FFR measurement only records the distal pressure at target vessel, resulting that the side branches could not be included in the CFD model (lacking of boundary conditions to include side branches). This simplification was likely to have a significant effect on the computed flow rate. According to the scaling laws in a vascular tree, the resistance of each vessel branch inversely related to the radius [21]. Thus, ignoring the side branches would overestimate the flow rate and how much was overestimated depends on the relative size of the ignored side branches. Recently, Gosling et al. proposed a leakage model to compensate the impact of unresolved side-branches in CFD models [22]. In the leakage model, artery was simulated as a “porous” wall, through which fluid flowed at a rate determined by local geometry. Taylor et al. further validated this model against thermodilution method [23]. However, the leakage model assumed that vessel taper corresponded to flow to side branches, which might not be true in diseased vessels. Moreover, lesions tend to occur at the bifurcation of coronary blood vessels [24], and would lead to complex flow patterns, which cannot be modeled in the leakage model.

Combining coronary angiography images and routine FFR measurement to assess the intracoronary flow rate is a problem of simulating blood flow under incomplete pressure boundary conditions. Fortunately, blood vessels can sense the shear stress on the endothelial cells and modulate their size to maintain the shear stress at a constant level, resulting in a form-function relationship between the size of vessel and the flow it carried [25,26]. In this paper, this form-function relationship was employed and a pressure-flow curve based flow resistance model was used to model resistance of the epicardial arteries, enabling to calculate the mean intracoronary blood flow rate in two or more branches under incomplete pressure boundary conditions (with only recording the proximal and distal pressure at target branch). To validate the proposed method, simulation experiments were performed since it was difficult to obtain the “golden standard” of intracoronary flow rate *in vivo*. The incomplete boundary conditions that would, in clinical practice, be measured by FFR, were simulated by using pulsating blood flow simulation method [27]. Then, the intracoronary flow rates computed by the proposed method could be validated against the simulated results.

2. Methods

2.1. CFD based computational method

The pressure drop of the epicardial coronary artery can be approximately modeled by a common fluid dynamic equation:

$$P_a - P_d = VF \cdot \bar{Q} + EL \cdot \bar{Q}^2 \quad (1)$$

where P_a is the mean proximal pressure, P_d is the mean distal pressure, $\bar{Q}$ is the mean flow rate, VF and EL are flow resistance parameters of epicardial coronary artery, namely viscous friction (VF) and expansion loss (EL) [28]. These two flow resistance parameters (VF and EL) are mainly determined by the anatomy of epicardial coronary artery. P_a and P_d at the target vessel can be measured by the routine FFR measurement, thus if we obtain the VF and EL of the target vessel, then the $\bar{Q}$ can be calculated from Eq. (1).

With a given 3D model of coronary vessel, steady flow simulations were performed three times with different flow rates. Each time the simulated mean flow rate and mean pressure drop at the target vessel were extracted. These data combined to provide three points in pressure-flow coordinate system; then VF and EL were estimated by using iterative least squares estimation for nonlinear regression [29,30]. The specific details of the steady flow simulations were as follows: Inlet was applied a static flow rate boundary condition, and was set to be q , $2q$ and

$3q$ each time. Outlet boundary conditions were kept the same for the three times of steady flow simulations: lumped parameter models with only one resistance were coupled at each outlet. Total distal resistance was specified as R_0 , and was allocated to each outlet according to the Murray's Law [21]:

$$R_m \propto \frac{1}{r_m^k} \quad (2)$$

Where R_m was the resistance at each outlet, r_m was the radius of the branch vessel, k was the coefficient of the Murray's Law (set to 2.67 for coronary vessel). The two parameters, q and R_0 , were chosen to corresponding to a normal physiological range, and were set to be 0.5 ml/s and 80 000 dynes s/cm⁵, respectively. SimVascular, an opensource software package was used perform the CFD simulation [31]. The blood was modeled as an incompressible Newtonian fluid. The dynamic viscosity was set to be 4.0 cP and the density to be 1060 kg/m³. The simulated mean flow rates and mean pressure drops at the target vessel were exported to MATLAB R2019a (The MathWorks, Inc., Natick, MA, USA) for fitting the VF and EL , and then combined with the measured mean pressure drop, the mean intracoronary flow rate was calculated from Eq. (1). The proposed model can be used in clinical practice to calculate the intracoronary flow rate from coronary angiography images and routine FFR measurement, and the flowchart of how the proposed method would be used in clinical practice was illustrated in Fig. 1.

2.2. Simulation experiments

Because it was difficult to obtain the “golden standard” of intracoronary flow rate *in vivo*, simulation experiments were designed to validate the proposed method. Both idealized and reconstructed 3D models of coronary branches were employed; Pulsating blood flow in these 3D coronary models was simulated by using CFD method, then the pressure distribution and the corresponding flow rate were obtained for each 3D model; Finally, the simulated mean pressure at inlet and outlet were extracted to model the incompletely boundary conditions measured by FFR, and the simulated corresponding flow rate was extracted as the reference of the intracoronary flow rate. The flowchart

of the simulation experiments was illustrated in Fig. 2 and the details were as follows:

2.2.1. 3D models of coronary branches

Idealized 3D model: A series of Y-shaped 3D models were constructed, as shown in Fig. 2(a). The radius of the target branch was set to be 1.2 mm. Radius ratio (r ratio) was defined as the radius of the side branch divided by the radius of the target branch. Focal lesion with a length of 6 mm was insert in the target branch, and the diameter stenosis ($DS\%$) was defined as the 100% minus the percentage of minimal lesion diameter to the normal lumen diameter. We varied the r -ratio from 0.5 to 1.0 (0.5, 0.6, 0.7, 0.8, 0.9, 1.0) and the $DS\%$ from 50% to 75% (50%, 55%, 60%, 65%, 70%, 75%), generating a total of 36 3D Y-shape models.

Reconstructed 3D model: Patients (aged at least 18 years), who referred for diagnostic coronary angiography in the Department of Cardiology, Xijing Hospital, Fourth Military Medical University between November 2020 and October 2021 due to stable or unstable angina were investigated. This study was approved by the Ethics Committee of Fourth Military Medical University (KY20202042-C-1, Xi'an, China). Written informed consent was obtained from each participant. Coronary angiography images from multiple views were recorded at 15 frames per second. The DICOM images were exported to the Northwestern Polytechnical University for analysis. 3D centerlines of the targeted vessel and its large branches (diameter > 1 mm and length > 2 cm) were reconstructed from two angiographic projections (separated by $\geq 25^\circ$) by using the point-cloud based approach [32], then circular luminal contours were fitted along all vessel paths to generate the 3D anatomy models. An example of the reconstructed model was shown in Fig. 2(b).

2.2.2. Pulsating blood flow simulation

Pulsatile volume inflow profiles were applied as velocity inlet boundary conditions. Flow rate was normalized and then set to 1 ml/sec. The Windkessel model was coupled to each outlet, and the total resistance and capacitance were set to be 62,160 dynes s/cm⁵ and 10^{-6} cm⁵/dynes. Ratio of the two resistances of the Windkessel model for each outlet was 1:9. The total resistance or capacitance were split according

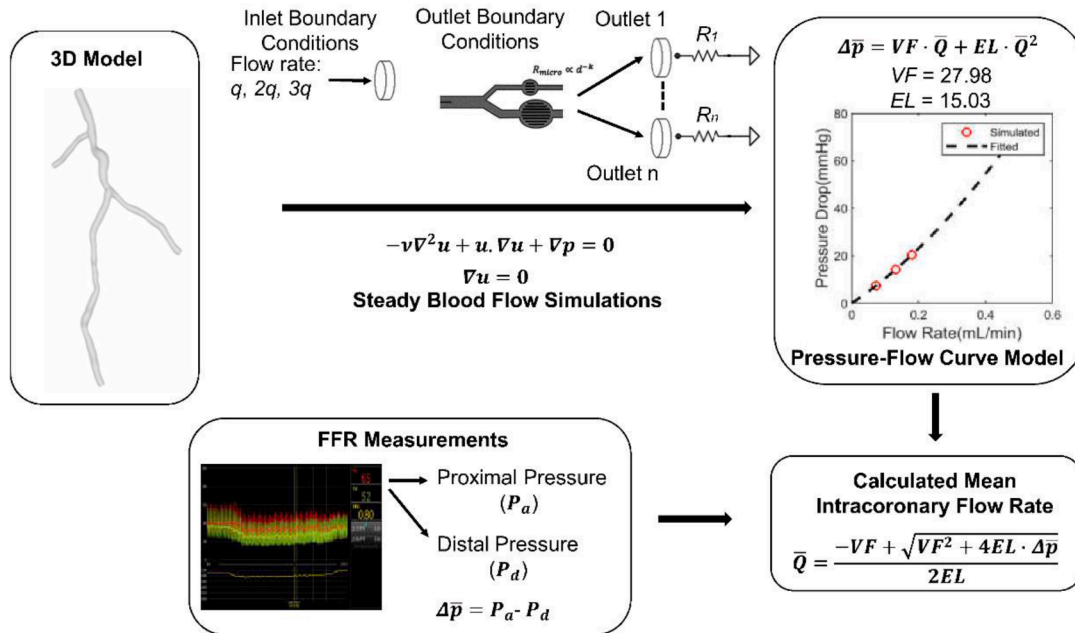


Fig. 1. Flowchart of how the proposed method would be used in clinical practice. With a given 3D model of coronary vessel, steady flow simulations were performed three times with different flow rates; then, the simulated mean flow rate and mean pressure drop at the target vessel were extracted and parameters of the pressure-flow curve model, VF and EL , were fitted; finally, with the incomplete pressure boundary conditions obtained by FFR, the mean intracoronary flow rate was calculated.

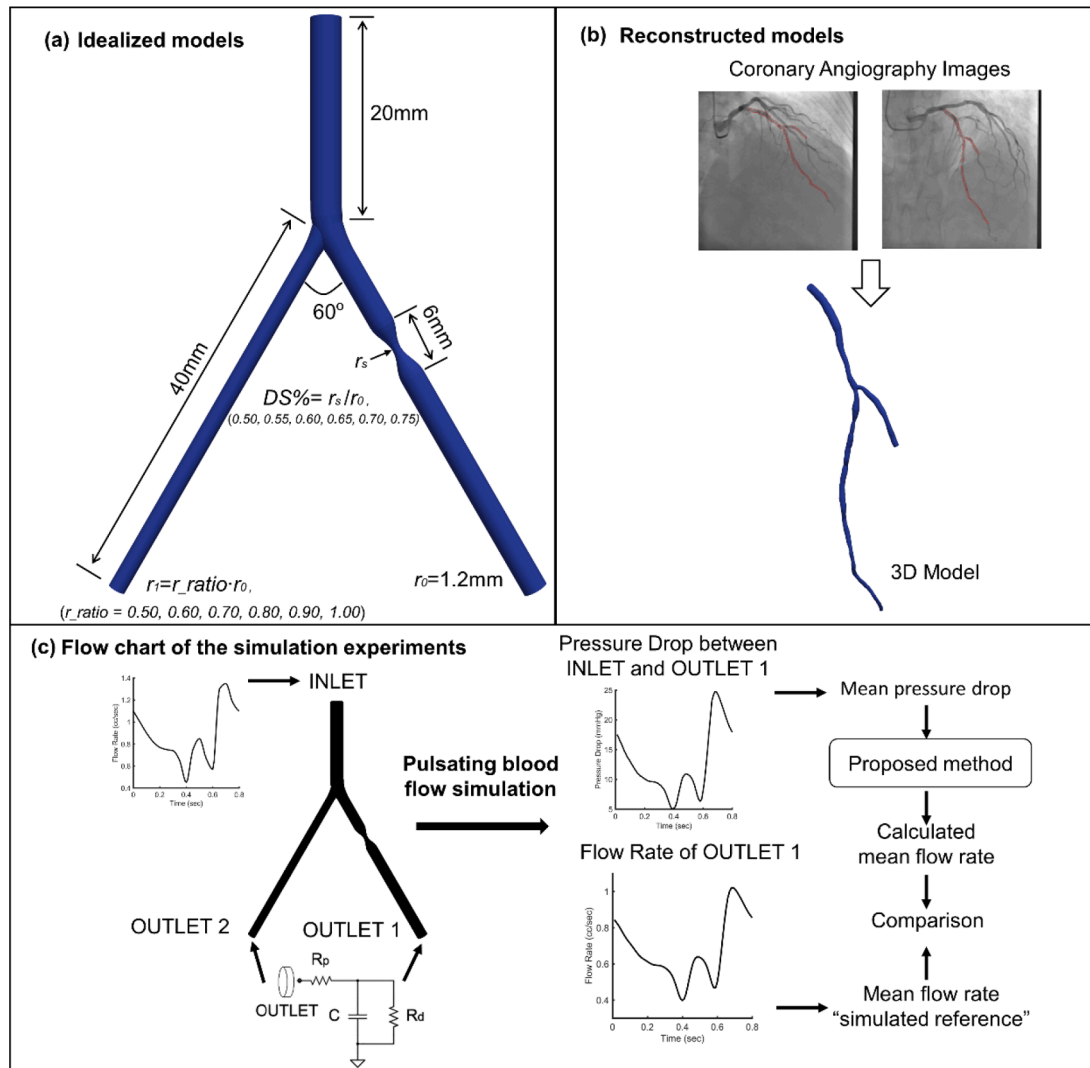


Fig. 2. 3D models and the flowchart of the simulation experiments. (a) Idealized Y-shape 3D model: the r -ratio varied from 0.5 to 1.0 (0.5, 0.6, 0.7, 0.8, 0.9, 1.0) and the $DS\%$ varied from 50% to 75% (50%, 55%, 60%, 65%, 70%, 75%), generating a total of 36 3D Y-shape models. (b) An example of the reconstructed 3D model; (c) flowchart of the simulation experiments (to model the incomplete boundary conditions that would, in clinical practice, be measured by FFR): Pulsating blood flow in these 3D coronary models was simulated by using CFD method; then the pressure distribution and the corresponding flow rate were obtained for each 3D model; finally, the simulated mean pressure at inlet and outlet were extracted to model the incompletely boundary conditions measured by FFR, and the simulated corresponding flow rate was extracted as the "golden standard" of the intracoronary flow rate. (r -ratio: Radius ratio; $DS\%$: diameter stenosis.).

to the Murray's Law. The coefficient of the Murray's Law was first set to be 2.67, and then changed to 2.33 and 3.00 to investigate the performance of the proposed method when the assumed coefficient of the Murray's Law (2.67) was inconsistent with the real blood flow condition. Still, SimVascular was used to perform the CFD simulation. The solutions were run for three cardiac cycles and only the results of the third cardiac cycle were used.

The proposed approach was compared with two methods: (1) model without side branch (noSideBranch): without considering the side branch, the 3D model only contained the target branch. The mean pressure at proximal and distal section was directly applied at inlet and outlet [20]; (2) simple boundary condition model (simpleBCD): with the side branch, the outlets of each branch were assumed to have the same mean distal pressure (mean pressure at proximal section was applied at inlet).

3. Results

The calculated mean flow rate for the idealized models were shown in Fig. 3. The proposed method slightly overestimated (less than 0.023

ml/s) the mean flow rate as compared with the simulated reference for all models. While, overestimations of the mean flow rate were observed for the noSideBranch method and the simpleBCD method. As the $DS\%$ increase from 50% to 75%, the overestimations for those two compared methods increase significantly, and these phenomena were observed for models with all r ratio.

The percentage errors varied with the pressure drop and were shown in Fig. 4 for idealized models. The simulated pressure drops ranged from 2.01 mmHg to 74.75 mmHg. The mean percentage error was $+2.05\% \pm 0.04\%$ for the proposed method, and was much lower than that of the noSideBranch method ($+38.48\% \pm 10.45\%$) and the simpleBCD method ($+31.34\% \pm 9.27\%$). As the pressure drop increased, the percentage errors of the proposed method maintained below $+3.06\%$ for all models, but the percentage errors increased to $+120.66\%$ for the noSideBranch method and to $+112.06\%$ for the simpleBCD method.

Fig. 5 demonstrated the performance of the proposed method when the assumed coefficient of the Murray's Law (2.67) was inconsistent with the real blood flow condition (2.33 and 3.00). From the results, even the pulsating blood flow had a different coefficient of the Murray's Law, the percentage errors still maintained at a very low level (less than

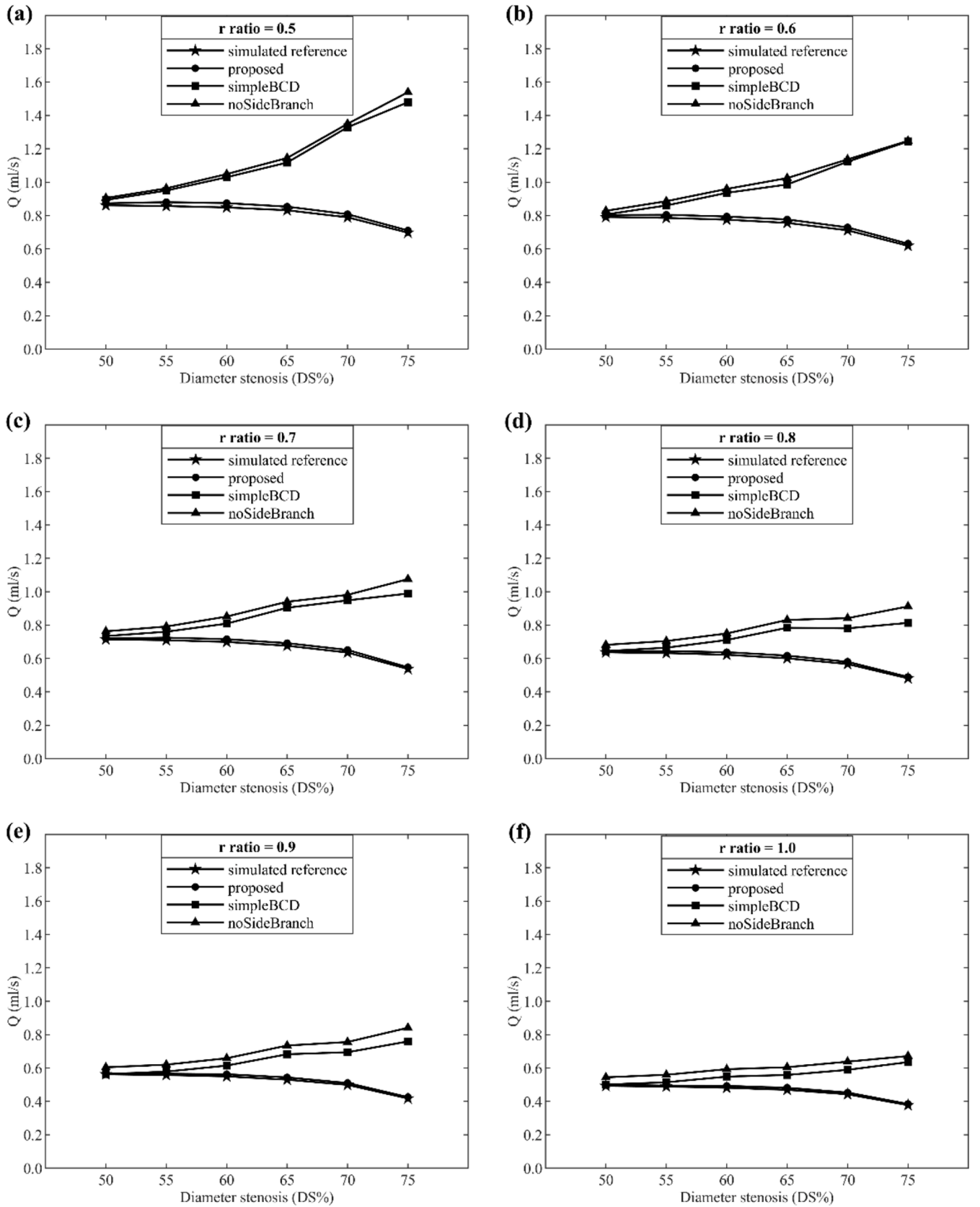


Fig. 3. Comparisons of the calculated mean flow rate for the idealized models. (a) models with r-ratio equal to 0.5; (b) models with r-ratio equal to 0.6; (c) models with r-ratio equal to 0.7; (d) models with r-ratio equal to 0.8; (e) models with r-ratio equal to 0.9; (f) models with r-ratio equal to 1.0. (r-ratio: Radius ratio; DS%: diameter stenosis.).

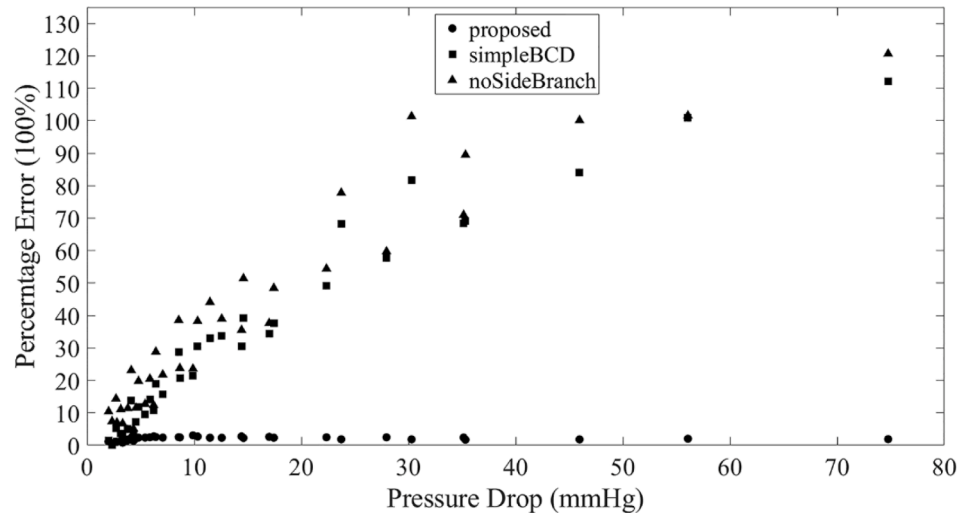


Fig. 4. Comparisons of the percentage errors for idealized models.

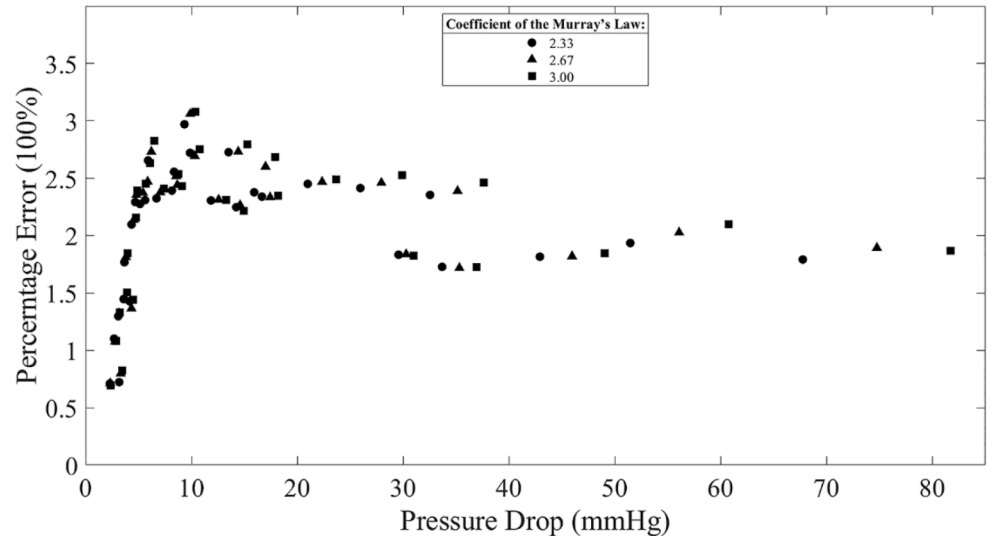


Fig. 5. The percentage errors of the proposed method when the assumed coefficient of the Murray's Law (2.67) was inconsistent with the real blood flow condition (2.33 and 3.00) for idealized models.

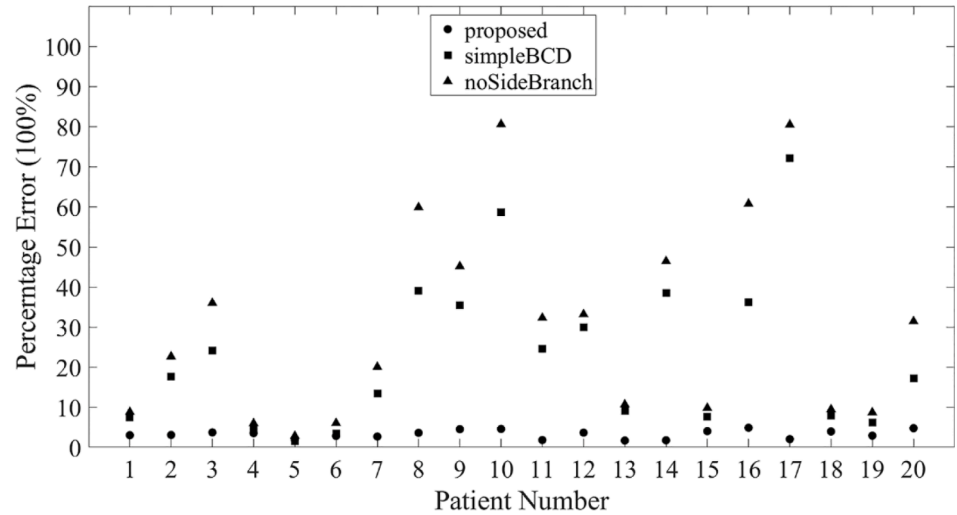


Fig. 6. Comparisons of the percentage errors for reconstructed models.

about 3.00%).

A total of 20 patient-specified 3D models were reconstructed from 20 Patients. The anatomy details of the 20 reconstructed models were shown in Supplementary Material (Table 1). Fig. 6 showed the percentage errors for 20 patient-specified 3D models. The results were similar to the idealized model. The mean percentage errors were $+2.24\% \pm 0.01\%$ for the proposed method, and was much lower than that of the noSideBranch method ($+30.54\% \pm 6.12\%$) and the simpleBCD method ($+22.74\% \pm 3.72\%$).

4. Discussion

Results demonstrated that side branches had great impact on the calculated blood flow rate. Ignoring the side branch would lead to overestimation of the flow rate, especially when a significant stenosis existed in the target vessel. Even when the side branch was much smaller than the target vessel ($r_{ratio}=0.5$), the calculated blood flow rate for the noSideBranch method overestimated 120.66% when a significant stenosis existed ($DS\%=75\%$). Indeed, how flow was divided into branches was determined by the flow resistance of each branch, which contained both flow resistance of epicardial arteries and distal micro-vessels [33]. During normal condition, the flow resistance of the epicardial arteries was much smaller than that of the distal micro-vessels, and the flow resistance of distal micro-vessels followed the “form-function” relationship [21,33]. However, significant stenosis would increase the resistance of the epicardial arteries, redistributing blood flow between branches. During this condition, even ignoring (or applying improper boundary conditions) a small branch would lead to huge error in the calculation of blood flow rate.

Previous experiment study demonstrated that pressure losses over a stenosis can be approximately modeled by Eq. (1) [28]. Our group demonstrated that coronary arteries with side branches also fit the Eq. (1) and proposed a CFD based method to extract the VF and EL from the 3D anatomy model of epicardial coronary artery [30]. In this paper, we changed the way of setting boundary conditions for the steady flow simulation to improve the convergence, and then applied this model to calculate of the intracoronary flow rate. Results showed that the proposed method had a good accuracy over a wide range of pressure drops. Coefficient of the Murray's Law was the key parameter involved in the proposed method. Theoretical derivations of turbulent flow suggested that the value of the Murray's Law might range from 2.33 to 3.0 [34]. Recent research using optimized CFD method suggested that 2.33 was more accuracy [23,35]. However, limited data was available for diseased arteries at hyperemia condition. Hutchins et al. reported that stenosis would have great impact on the value of the Murray's exponent [36]. These evidences suggested that the exactly value of Murray's exponent might vary from patient to patient, but in a range from about 2.33 to 3.0. In this work, we set it to be an intermediate value (2.67) and investigated the performance of the proposed method when the assumed coefficient of the Murray's Law was inconsistent with the real blood flow condition. From our results, this parameter had great impact on the real pressure distribution and flow rates, but fortunately, it had limited impact on the accuracy of the proposed method. Indeed, when the coefficient of the Murray's Law changed, the two model parameters (VF and EL) changed much less. Thus, the coefficient of the Murray's Law had limited impact on the relationships between the mean pressure drop and mean flow rate. This was why the proposed method had a good accuracy even when the assumed coefficient of the Murray's Law was inconsistent with that of the real blood flow condition.

In catheter laboratory, intracoronary pressure can be measured accurately with pressure wires. It becomes a routine measurement for assessing the significant hemodynamic stenosis [37]. However, the measurement of intracoronary flow is much more challenging. Though Doppler sensor or thermodilution technique provided ways to assess the intracoronary flow, the utilization was limited [12]. Except for the technical difficulties, the measured flow rates always need to be

combined with the measured pressure to provide complementary physiological information [38]. Thus, in clinical practice, dual-sensor wires (pressure and flow) were used when flow rates need to be measured. These dual-sensor wires greatly increased the complexity of the measurements and limiting its in-hospital utilization [12]. The proposed method provided an easier way to assess the intracoronary flow rate. Since it was derived from routine coronary angiography and FFR measurements, no additional equipments or operations were needed. We also demonstrated the flexibility of the proposed method in 20 patients, but currently we could not make a head-to-head comparison with Doppler sensor or thermodilution technique, thus these results were only provided in the supplementary materials.

There are several limitations in this study. First, the proposed method was validated through simulation experiments, and further in *vitro* and in *vivo* validation is needed, especially head-to-head comparisons with Doppler sensor or thermodilution technique. Second, the proposed method only calculated the mean intracoronary flow rate, and the turbulence for models with significant stenosis was ignored. Advanced CFD model needs to be further employed to compute the time varied intracoronary flow rate more accurately; Third, the unresolved side branches were ignored in the proposed method, and it may be improved by combining with the leakage model. Fourth, the anatomy of epicardial arteries changes during the cardiac cycle, but the 3D models could only be derived from a certain frame (diastole). Advanced method which takes the changes of the anatomy during cardiac cycle into consideration is needed.

5. Conclusions

In this paper, a novel CFD based method was proposed to calculate the mean intracoronary flow rate from 3D anatomy model and routine FFR measurement. A pressure-flow curve based flow resistance model was employed to model resistance of the epicardial arteries. A series of steady flow simulations were performed to extract the parameters of the flow resistance model, which implicitly specified constraints for splitting flow between branches and thus enable the mean intracoronary blood flow rate to be calculated in two or more branches with only recording the proximal and the distal pressure at target branch. Results demonstrated that the proposed method had a good accuracy within a wide range of physiological conditions. The proposed method provided an easy and accurate way to measure the mean intracoronary flow rate and would facilitate the accurate diagnosis of IHD.

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Ethical approval

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CRediT authorship contribution statement

Weihao Li: Investigation, Writing – original draft. **Kun Lian:** Investigation, Writing – original draft. **Yan Chen:** Methodology. **Shuai Zhao:** Methodology. **Xiaoqiong Guo:** Formal analysis. **Qian Tao:** Writing – review & editing. **Haokao Gao:** Formal analysis. **Songyun Xie:** Resources. **Chengxiang Li:** Resources. **Qiong Wang:** Conceptualization. **Xin Zhou Xie:** Supervision.

Declaration of Competing Interest

None declared

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None

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.medengphy.2022.103942](https://doi.org/10.1016/j.medengphy.2022.103942).

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